

THOMAS ROUTHU

PhD Candidate in Computer Science | Machine Learning | Deep Learning | Edge AI
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Professional Summary

PhD Candidate in Computer Science with 3+ years of experience spanning **machine learning, deep learning, data engineering, time-series analysis, sensor data, and industrial automation**. Experienced in designing data pipelines, processing large-scale multivariate sensor datasets, developing anomaly detection workflows, and translating complex data into actionable insights. Currently working on industrial AI applications involving sensor-based infrastructure monitoring and predictive failure analysis. Strong background in **Python, SQL, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, data visualization, ETL, data quality assurance, and workflow optimization**, with experience collaborating across academic and industrial environments.

Technical Skills

Programming: Python, SQL, MATLAB

Machine Learning & AI: Machine Learning, Deep Learning, Artificial Neural Networks, Anomaly Detection, Predictive Modeling, Scikit-learn, TensorFlow, PyTorch

Data Science & Analysis: Time-Series Analysis, Sensor Data Analysis, Data Preprocessing, Feature Engineering, Statistical Analysis, Data Visualization, NumPy, Pandas, Matplotlib, Seaborn

Data Engineering & Operations: Data Pipeline Design, ETL Processes, Data Quality Assurance, Data Annotation, Workflow Optimization, KPI Development

Tools & Technologies: Git, Docker, Linux, OpenCV, Excel, Tableau, Power BI, ANSYS, SolidWorks

Professional Experience

Visiting PhD Student

OWL University of Applied Sciences, Institute Industrial IT

August 2024 – Present

Lemgo, Germany

- Design and manage end-to-end data workflows for vibration and inclinometer measurements from **100+ sensors** across multiple bridge spans, ensuring data integrity and quality validation.
- Analyze large-scale **multivariate time-series datasets** using deep learning methods to identify patterns, anomalies, and potential failure events.
- Establish **KPIs and monitoring frameworks** to evaluate anomaly detection performance and support optimization of sensor deployment strategies.
- Develop data-driven approaches for intelligent maintenance and infrastructure monitoring applications.
- Collaborate with engineers and researchers to align data analysis strategies with infrastructure monitoring and smart bridge initiatives.

Industrial PhD – Computer Science / Machine Learning

Milano Serravalle – Milano Tangenziali S.p.A.

October 2023 – Present

Milan, Italy

- Develop and maintain data collection, preprocessing, and annotation workflows for **sensor-based infrastructure monitoring** using Deep Neural Networks and machine learning techniques.
- Process and analyze complex sensor datasets to identify **patterns, anomalies, and data-quality issues**, translating findings into data-driven recommendations.
- Develop reporting frameworks and KPIs to track model and workflow performance for infrastructure monitoring systems.
- Continuously improve data processing pipelines to increase **scalability, reliability, and data quality** across monitoring activities.
- Collaborate with technical and research stakeholders to develop machine learning solutions for intelligent infrastructure monitoring.

Samsung Innovation Campus

Samsung Italia

November 2023 – January 2024

Pavia, Italy

- Participated in an advanced industry training program focused on **Artificial Intelligence (AI), Internet of Things (IoT), and digital transformation**.
- Developed technical and professional skills relevant to **digital innovation, emerging technologies, and technology-driven transformation**.

Robotics Programmer – Working Student
ReHouseit – Società Benefit

October 2022 – September 2023
Milan, Italy

- Designed and optimized automated workflows for **industrial collaborative robot (cobot) operations**, contributing to process optimization and project execution.
- Developed specialized software tools for **CAM translation and real-time simulation**, improving operational efficiency by **30%**.
- Managed data collection and validation processes for cobot performance metrics, ensuring data accuracy and quality.
- Identified and resolved technical challenges through programming, troubleshooting, and collaboration with engineering teams.

Research Associate
BML Munjal University

January 2020 – June 2020
Gurgaon, India

- Conducted data-driven research on **tool coating optimization**, analyzing experimental datasets to improve tool-life performance by **25%**.
- Collected, processed, and analyzed materials-testing data and translated experimental results into research findings.
- Contributed to a **peer-reviewed international journal publication** based on experimental research.
- Maintained high standards of experimental data quality, documentation, and analysis.

Education

PhD in Computer Science – Data Science & Machine Learning Focus
Università di Pavia

October 2023 – 2026
Pavia, Italy

Master’s Degree in Robotics and Mechatronics – Industrial Automation
Università di Pavia

November 2020 – October 2023
Pavia, Italy

Master of Science in Informatics
Gdańsk University of Technology

March 2022 – September 2022
Gdańsk, Poland

Bachelor of Technology in Mechanical Engineering
BML Munjal University

August 2016 – May 2020
Gurgaon, India

Certifications & Professional Coursework

- **Machine Learning Specialization** – Stanford
- **Structuring Machine Learning Projects** – Stanford
- **Feature Engineering & Intermediate Machine Learning** – Kaggle
- **OpenCV Bootcamp – Computer Vision Applications**
- **Build Three Real-World Python Applications** – Applied Programming

Selected Publications

- **Modular Digital Twin for Air Handling Units** – Research on data-driven digital twin systems.
- **Efficient Storage by Filtering of Large Sensor Data for Infrastructure Monitoring** – Data optimization for large-scale sensor datasets.
- **Applying AI in the Area of Automation Systems: Overview and Challenges** – AI/ML applications in automation systems.
- **Probabilistic Framework for Multi-Vehicle Tracking Using Distributed Accelerometer Sensors** – AI/ML applications in automation systems.